Apr 13 2021

Verilog Weekly Quiz 1 -

20	10	10	0	50%	09:12
Questions	Right	Wrong	Unattended	Accuracy	Time Taken

10
Marks Scored

OFFER

View rank list (https://elearn.maven-silicon.com/report/assessment/998)

Topics	No.of Qns	Right	Wrong	Unattended	Accuracy	Remarks
BA	1	1	0	0	100%	Excellent
Process	3	2	1	0	66.67%	Average

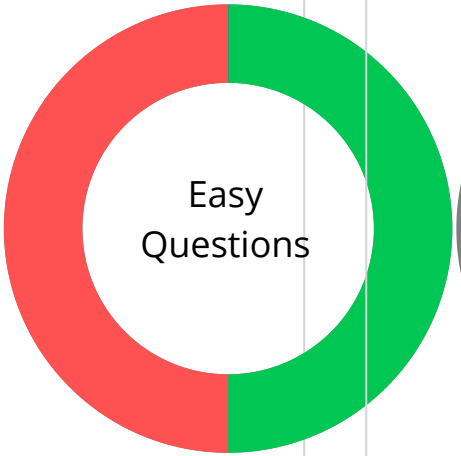
OFFER

Topics	No.of Qns	Right	Wrong	Unattended	Accuracy	Remarks
Synthesis	1	0	1	0	0%	Needs Improvement
Race	1	0	1	0	0%	Needs Improvement
Function	1	0	1	0	0%	Needs Improvement
Data-types	2	1	1	0	50%	Needs Improvement
Compiler directive	2	2	0	0	100%	Excellent
System function	1	0	1	0	0%	Needs Improvement
Operators	1	1	0	0	100%	Excellent

Topics	No.of Qns	Right	Wrong	Unattended	Accuracy	Remarks
Delays	1	0	1	0	0%	Needs Improvement
System task	2	2	0	0	100%	Excellent
Testbench	1	1	0	0	100%	Excellent
Task	1	0	1	0	0%	Needs Improvement
NBA	1	0	1	0	0%	Needs Improvement
Branch	1	0	1	0	0%	Needs Improvement

OFFER

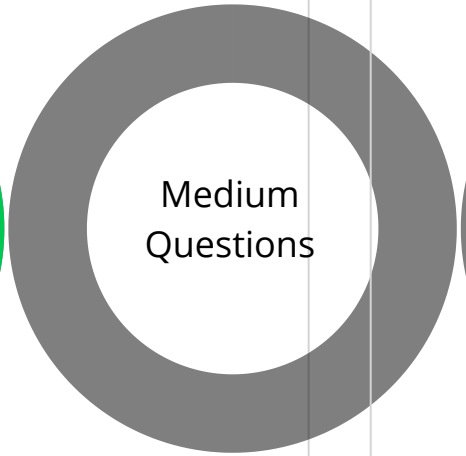
OFFER



Right Answers 10

Wrong Answers 10

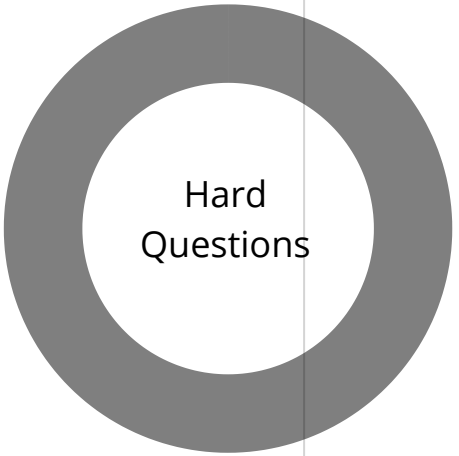
Unattended 0



Right Answers 0

Wrong Answers 0

Unattended 0



Right Answers 0

Wrong Answers 0

Unattended 0

All questions

Compare (https://elearn.maven-silicon.com/report/assessment/998)

1 Single choice

Find out the final value of out2 at every pos-edge of the clock and choose right answer?

OFFER

```
module ab(clk,in1,in2,out2);
input clk,in1,in2;
output out2;
reg temp,temp1,temp;

always@(posedge clk)
begin
temp = (in1^in2);
temp1= temp;
temp2 = temp1;
end

assign out2=temp2;
endmodule
```

a) **Previous value of temp1**

b) **Previous value of temp2**

c) **in1 ^ in2**

d) **concurrent execution (un predictable)**

e) **compilation error**

Your answer C **Right** Time Taken **48s** **Marks 1** **Answer Explanation**

Blocking updates are immediate assignments.

OFFER

2 Single choice

What is the reset type from the below Verilog snippet?

```
always@(posedge clock)
  begin
     if(~reset)
       q <= 0;
  end
```

a) Active high synchronous re set

b) Active low synchronous re set

c) Active high asynchronous clear

d) Active low asynchronous clear

e) None of the above

Your answer (b) **Right** Time Taken **8s** **Marks 1** **Answer Explanation**

Since the design resets wrt posedge clock only hence the reset is synchronous.

OFFER

3 Single choice

For combinational circuit synthesis, which of the following are recommended?

a) Feedback connections from output to input may be allowed.

b) If "if…else" construct is used, value must be assigned to the output variable for all possible combinations of the inputs.

c) Procedural assignment must not be used.

d) All of the above.

e) None of the above

Your answer (d) **Wrong** Right answer (b)

Time Taken **31s** **Marks 0** **Answer Explanation**

Feedback connection from output to input results in sequential behavior and must not be present in a combinational circuit. Thus option (a) is correct. If values of output vari ables are not specified for all input permutation, the circuit may be realized sequential latch. Thus option (b) is also correct. Continuous as well as procedural assignment bot h can be used in realizing combinational circuit. Thus option (c) is not correct.

OFFER

4 Single choice

What will the following code segment do?

```
initial
begin int1=42; int2=17; end
always @(posedge clock)
int2 = int1;
always @(posedge clock)
int1 = int2;
```

a) A race condition will occur, i.e. both the integer variables will update with either value of “int1” or “int2”.

b) Swaps the values of the variables “int1” and “int2”.

c) Both variables will get updated with the value 42.

d) Both variables will get updated with the value 17.

e) None of these

Your answer (b) **Wrong** Right answer (a)

Time Taken **43s** **Marks 0** **Answer Explanation**

Both the variable int1 and int2 will be updated with the same value of int1 or int2 based on order of execution of the two procedural blocks, which is a race condition. Thus option (a) is true and option (b), (c) and (d) are false.

5 Multiple choice

Which is a valid bug in the below code?

OFFER

```
reg [1:0]y ;

  function calc(input [1:0]a,b);
      begin
        calc = a + b ;
     end
  endfunction

 always@*
  y = calc(a,b);
```

OFFER

a) Function should have outp ut argument to return the su m output.

b) The width of the function s hould be 2bits wide.

c) The width of the function s hould be 3bits wide.

d) The width of the function & Size of the variable “y” should be 3bits wide.

e) None of the above

Your answer (a) **Wrong** Right answer (d)

Time Taken **66s** **Marks 0** **Answer Explanation**

NA

6 Single choice

What is the correct syntax for declaring a real number?

a) real r[1:0];

b) real r;

c) real [1:0] r;

d) All of the above

e) None of above

Your answer (b) **Right** Time Taken **14s** **Marks 1** **Answer Explanation**

NA

7 Single choice

Which of the following is used to customize simulation time?

a) `ifdef

b) `timescale

c) `define

d) delays

e) None of above

OFFER

Your answer (b) **Right** Time Taken **16s** **Marks 1** **Answer Explanation**

Timescale has Time-unit and Time-precision which is used for customization.

8 Single choice

Which of the following is also called as conditional compiler directive?

a) `ifdef

b) $test$plusargs

c) if else

d) All of the above

e) None of above

Your answer (a) **Right** Time Taken **14s** **Marks 1** **Answer Explanation**

`ifdef returns either True or False as per the macro definition.

OFFER

9 Single choice

Which system function returns a 64bit simulation time?

a) $realtime

b) $time

c) $stime

d) All of the above

e) None of above

Your answer (d) **Wrong** Right answer (b)

Time Taken **11s** **Marks 0** **Answer Explanation**

NA

10 Single choice

What should be the data-type for the net a in the below snippet such that it is driven with a value 1 only if all the 3 drivers “b, c, d” are driving 1.

OFFER

```
assign a = b;

assign a = c;

assign a = d;
```

a) wor

b) trireg

c) wand

d) All of the above

e) None of above

Your answer (a) **Wrong** Right answer (c)

Time Taken **37s** **Marks 0** **Answer Explanation**

Wired and is used in continuous concurrent assignments in multiple parallel drivers driving common node. The wired output goes 1 if all the drivers are driving 1.

11 Single choice

OFFER

What has to be done in the below code in order to get the shifted output as 4'b1100? In the existing code the output is evaluated as 4'b0100.

```
reg [3:0]a,b,sum;

   initial
     begin
      a = 4'b1100;b=4'b1101;
      sum = (a+b) >> 1;
      #10;
     end
```

a) sum = a + (b >> 1);

b) sum = a+b;

c) sum = (a+b+0)>>1;

d) All of the above

e) None of above

Your answer **C** **Right** Time Taken **17s** **Marks 1** **Answer Explanation**

a+b+0 converts the expression length to 32bit length hence the carry bit is preserved and w hen shifted by 1 towards right results in 4'b1100.

OFFER

12 Single choice

What is the output displayed from the below snippet?

```
integer i,j;
  initial
  fork
    for(i=0;i<10;i=i+1)
      $display("a=%d",i);
    for(j=0;j<10;j=j+1)
      $display("b=%d",j);
  join
```

a)

```
a=0
b=0
a=1
b=1
....
```

b) Depends on simulator hence output can be A or C

OFFER

```
a=              0
a=              1
a=              2
a=              3
a=              4
a=              5
......
b=              0
b=              1
b=              2
b=              3
b=              4
b=              5
......
```

c)

d) Can’t predict

e) None of above

Your answer (d) **Wrong** Right answer (b)

Time Taken **49s** **Marks 0** **Answer Explanation**

When you have processes executing in parallel, there is no deterministic ordering between s tatements executing at the same time. Both are valid possibilities. Another simulator might have picked the second for-loop to execute first.

13 Single choice

W hat is the name of the delay used in the below assignment?

```
wire #1 c = a1 | b1;
```

a) Inertial delay

b) Regular delay

c) Net delay

d) Inter delay

e) None of above

Your answer (b) **Wrong** Right answer (a)

Time Taken **16s** **Marks 0** **Answer Explanation**

Inertial delay acts like gate component delay.

14 Single choice

What is the display output in the following snippet?

OFFER

OFFER

```
reg [1:0]a,b;

initial
  begin
    a <= 2'd3;
    b <= 2'd2;
    #10;
    a <= 2'd0;
    b <= 2'd1;
    $monitor("Values of a=%d \t b=%d",a,b);
  end
```

a)

```
Values of a=3    b=2
Values of a=0    b=1
```

b)

```
Values of a=0    b=1
```

c)

```
Values of a=x    b=x
Values of a=3    b=2
Values of a=0    b=1
```

d)

```
Values of a=0    b=1
Values of a=3    b=2
```

e) None of the above

Your answer (b) **Right** Time Taken **17s** **Marks 1** **Answer Explanation**

$monitor is called at 10time unit hence will be monitoring the events from current time slot i.e 10ns.

15 Single choice

What is true for the following snippet?

```
always@*
  if(sel)
    y = a;
```

a) It infers a MUX

b) It infers a transparent latch

c) always@* is not a valid syntax

d) Its non-synthesizable

e) None of the above

Your answer (b) **Right** Time Taken **15s** **Marks 1** **Answer Explanation**

A latch is inferred since else part is missing. iytyuitr

OFFER

16 Single choice

Which is a valid syntax for $timeformat?

a) $timeformat("String to be displayed");

b) $timeformat($time);

c) $timeformat(time_unit,precision_num,suffix_string,minimum_field width);

d) $timeformat($realtime);

e) None of the above

Your answer C **Right** Time Taken **25s** **Marks 1** **Answer Explanation**

$timeformat is a system task used to set a uniform time for all sub-modules taking part in simulation in the display console.

17 Single choice

OFFER

Which is true for a self-checking testbench?

a) A self-checking Test-Benc h checks expected results a gainst actual results (DUT) obtained from the simulatio n.

b) A self-checking Test-Benc h is not allowed in Verilog HDL simulation.

c) A self-checking Test-Benc h is similar to Linear task b ased Testbench.

d) There is no such concept of Self-checking Testbench.

e) None of the above

Your answer (a) **Right** Time Taken **24s** **Marks 1** **Answer Explanation**

Self-checking Testbenches are testbenches used to compare actual DUT simulation results with expected values.

18 Single choice

Which of the following is syntactically correct in Verilog HDL?

OFFER

```
task static t;
  begin
    ...........
  end
endtask
```

a)

```
automatic task  t;
  begin
    ...........
  end
endtask
```

b)

```
task automatic t;
  begin
    ...........
  end
endtask
```

c)

```
function void ( );
  begin
    ...........
  end
endtask
```

d)

e) None of the above

Your answer (b) **Wrong** Right answer (c)

Time Taken **14s** **Marks 0** **Answer Explanation**

Automatic tasks are supported in Verilog HDL with automatic keyword before task identifier and after task keyword.

19 Single choice

Which of the following are defined regions in a Verilog stratified event queue?

a) Preponed->Active->Inactive->NBA->Postponed

b) Preponed->Active->Inactive->NBA->Reactive->Postponed

c) Active->Inactive->NBA->Postponed

d) Active->Inactive->NBA->Reactive->Postponed

e) None of the above

Your answer (b) **Wrong** Right answer (c)

Time Taken **26s** **Marks 0** **Answer Explanation**

Verilog has 4 regions which are static: Active->Inactive->NBA->Postponed

OFFER

20 Single choice

What changes have to be done in the top module in-order to display “I am b” conditionally?

```
module a;
 initial
  $display("I am a");
endmodule

module b;
   initial
    $display("I am b");
endmodule


module top;

    parameter POWER = 1;

     ..........

endmodule
```

OFFER

OFFER

```
module top;
    parameter POWER = 1;
       generate
         if (POWER > 2)
            begin
              a a1();
            end
         else
             begin
               b b1();
             end
      endgenerate
endmodule
```

a)

```
module top;
    parameter POWER = 1;

      initial
         if (POWER > 2)
            begin
              a a1();
            end
         else
             begin
               b b1();
             end

endmodule
```

b)

```
module top;
    parameter POWER = 1;

        if (POWER > 2)
             begin
               a a1();
             end
          else
             begin
               b b1();
             end

endmodule
```

c)

```
module top;

  `ifdef POWER
     a a1 ( ) ;
  `else
    `ifdef
     b b1 ( );
  `endif
```

d)

e) None of the above

Your answer (e) **Wrong** Right answer (a)

Time Taken **49s** **Marks 0** **Answer Explanation**

Generate if-else can be used to instantiate module conditionally.

OFFER

Address

MAVEN SILICON
21/1A, III Floor, Marudhar Avenue,
Gottigere, Uttarahalli Hobli,
South Taluk,
Bannerghatta Road, Bangalore - 560076

Quick links

Terms and Conditions (https://elearn.maven-silicon.com/policies)
About Us (https://www.maven-silicon.com/about-us)
FAQs (https://elearn.maven-silicon.com/faqs)
Free VLSI Workshop for Freshers (https://elearn.maven-silicon.com/free-vlsi-workshop-freshers)
Privacy Policy (https://elearn.maven-silicon.com/privacy-policy)
VLSI Internship (https://www.maven-silicon.com/vlsi-design-internship)
Enquiry Form (https://elearn.maven-silicon.com/enquiry-form)

Contact Us

Call Us +91 9513398555
Email elearn@maven-silicon.com

(https://www.facebook.com/Maven-Silicon-326974926639)
(https://twitter.com/MavenSilicon)
(https://www.youtube.com/channel/UCu19H6thv_ad_Sl2cG-t-jA)

Copyright © 2021 Maven Silicon - Online VLSI Courses

OFFER